
Battery-Free, BLE-Enabled Wearable Device for Multi-Floor Indoor Emergency Alert Tracking

Team broshisis | ESW Lab 2

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Extend a validated, battery-free BLE wearable into a scalable multi-floor emergency alert system, from codebase and hardware to ML-based localization and real-world deployment.

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Problem Statement

Objective

Build upon the existing prototype to make it ready for real world deployment and add features to improve the user experience.

Project requirements

- Understand the existing wearable and receiver codebase.
- Design a better enclosure for the hardware
- Create a proper dashboard for visualisation of alerts generated by users
- Achieve reliable room and floor-level localisation indoors.

Example capabilities

- 01** Detect and localise an emergency press to a specific room or floor by fine tuning an ml model
- 02** Confirm that the signal is recieved with a live status LED and dashboard update
- 03** Scaling up the project to support the use of multiple wearables at the same time

Motivation

Fixed & battery-dependent

Battery-powered wearables need recharging, which makes them not suitable for emergency alert systems.

Battery-free and ML localisation

Using the energy harvesting system does away with battery upkeep altogether, and pairing this with RSSI fingerprinting and the ML model takes localisation to room-level accuracy.

Why this system?

Our mentor's prototype is already validated and piloted in a few rooms. We aim to scale it with better design, real data, and a live deployment.

KEY CHALLENGE Deliver an alert system that is reliable, precisely localized, and simple enough for a single press to trigger, even for non-verbal or limited-mobility users.

Zero maintenance

Accessible for all users

Precise localization

Our proposed solution



Localisation approach

- **RSSI fingerprinting**
Combine signal strength across all receivers into a single location fingerprint.
- **ML classification**
Train a classifier on labeled fingerprints to predict room and floor of origin.
- **Real-world validation**
Test across Himalaya before the special-needs school deployment.

System build

- **MQTT pipeline (existing)**
Receivers already publish RSSI readings via MQTT to a central broker.
- **Custom enclosures**
Durable, wearable-friendly housing plus a mountable receiver enclosure.
- **Status LED**
Blinks when a signal is received from a wearable.

Example outputs



Plan of Action

01 Codebase study

Understand existing codebase, firmware, MQTT pipeline, and RSSI logging.

02 Enclosure design

Build wearable and receiver enclosures.

03 Adding Features

Add status LED and other features to improve user experience.

04 Dashboard Creation

Create a proper centralised dashboard for visualising the received alerts using MQTT.

05 Deployment & data collection

Deploy across Himalaya block classrooms and corridors and collect data.

06 ML model training

Train and evaluate an RSSI fingerprint classifier or use the existing model.

07 Onsite testing

Deploy the full system at the special needs school.

08 Evaluation

Assess accuracy, reliability, and staff usability feedback.

EXPECTED DELIVERABLE A deployed, battery-free BLE alert system that localizes emergency presses to a specific room or floor in real time, validated onsite at a special needs school.